

IC-272 — Introduction to Machine Learning

Lab 4 — Linear Regression from Scratch

Indian Institute of Technology Mandi

Total marks: 100 — Duration: 2 hours

Marks: 60 for code, 40 for the viva. Coding marks are shown against each sub-part: Task 1 — 12, Task 2 — 18, Task 3 — 20, Task 4 — 10. The remaining 40 marks are awarded by your TA in a viva on your executed notebook, 10 marks per task. Be ready to explain any number your notebook prints and any plot it draws.

Rules

- Permitted: `numpy`, `pandas`, `matplotlib`. `scikit-learn` is **not permitted** — a task that calls it scores zero for that task.
- Barred for fitting: `np.polyfit`, `np.linalg.lstsq` and `np.linalg.pinv`.
- Allowed: `np.linalg.solve` and `np.linalg.inv` — assemble the system yourself.
- Use `np.random.default_rng(0)` wherever randomness is needed.
- Every function named in this handout must appear with exactly that name.

Dataset

`ic272_lab4_rent.csv` — 300 rental listings, five features and one target. Find the range of each column yourself as part of Task 1.

Column	Meaning
<code>area</code>	floor area (hundreds of sq. ft)
<code>rooms</code>	number of rooms
<code>age</code>	age of the building (units of 5 years)
<code>dist_km</code>	distance to campus (km)
<code>floors</code>	number of floors
<code>rent</code>	monthly rent (thousands of rupees) — target

The features are already on comparable numeric ranges, so no transformation is needed in this lab.

Demonstration — fitting a line by hand (TA, ≈ 15 min)

The TA works through this on the board and in a notebook. Nothing here is graded; it exists so that the formula in Task 1 is not a formula you have never seen before.

```
x = np.array([5., 6., 7., 8., 9.])      # area
y = np.array([35., 42., 46., 55., 60.]) # rent
```

Step 1. Plot the five points. They lie roughly on a line. The whole of linear regression is the question: *which* line?

Step 2. Guess one. Take $m = 5$, $c = 10$, i.e. $\hat{y} = 5x + 10$. Draw it. Predict a single point: at $x = 7$ the model says 45, the data says 46, so the error is 1.

Step 3. Score the guess. Define $SSE = \sum_i (y_i - \hat{y}_i)^2$ and compute it for this line: 55.0. A guess is now a number, and numbers can be compared.

Step 4. Guess better. Take $m = 6$, $c = 5$. The SSE drops to 7.0. Ask the room for a third guess and score it too. Fitting is nothing more than searching for the (m, c) that makes this number smallest.

Step 5. Stop guessing. The least-squares formulae give $m = 6.3$, $c = 3.5$ in one step, with $SSE = 4.3$. Invite the class to beat 4.3 by hand; they cannot, because no line has a lower SSE.

Close the demonstration by setting up the rest of the lab. With one feature you can eyeball a line; with five you cannot, so Task 2 solves for all six coefficients at once with linear algebra. When the matrix is too large even for that, Task 3 goes back to guessing — but lets calculus choose each next guess instead of the room.

Task 1 — Simple linear regression, closed form [12 marks]

- (a) Load the CSV with `pandas` and print the minimum and maximum of every column. [2 marks]
- (b) Write `train_test_split_scratch(X, y, test_frac, seed)` that shuffles the row indices with `np.random.default_rng(seed).permutation` and returns `X_train`, `X_test`, `y_train`, `y_test`. Take the `first int(n * test_frac)` shuffled indices as the test set and the rest as the training set — everyone must use this convention, or no two notebooks will agree. Call it with `test_frac=0.2`, `seed=0`. No library split helper. [5 marks]
- (c) Write `fit_simple_lr(x, y)` returning (w, b) from

$$w = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad b = \bar{y} - w\bar{x},$$

fit it on the training set with `area` as the only feature, and print w and b to three decimals. [3 marks]

- (d) Scatter `area` against `rent` for the test set with your fitted line drawn over it. Label both axes. [2 marks]

Task 2 — Multiple linear regression, normal equation [18 marks]

- (a) Write `design_matrix(X)` that prepends a column of ones to an (n, d) feature matrix and returns an $(n, d + 1)$ array. [4 marks]
- (b) Write `fit_normal_equation(X, y)` returning $\hat{\theta} = (A^\top A)^{-1} A^\top y$, where A is the design matrix. Use `np.linalg.solve` rather than forming the inverse, and say in a comment why. [9 marks]
- (c) Fit on all five features of the training set and print $\hat{\theta}$ to three decimals as a labelled table, one row per feature plus the intercept. [5 marks]

Task 3 — Batch gradient descent [20 marks]

For $J(\theta) = \frac{1}{n} \|A\theta - y\|^2$ with $\nabla J(\theta) = \frac{2}{n} A^\top (A\theta - y)$:

- (a) Write `fit_gradient_descent(X, y, lr, n_iters)` starting from $\theta = \mathbf{0}$, applying $\theta \leftarrow \theta - \eta \nabla J(\theta)$, and returning the final θ together with the list of $J(\theta)$ values. [10 marks]

- (b) Run it on the five-feature training set with `lr=0.005`, `n_iters=20000`. Plot the loss curve. Print your final θ beside $\hat{\theta}$ from Task 2 and report the largest absolute difference between them. [5 marks]
- (c) Re-run with `lr` $\in \{0.001, 0.005, 0.01\}$ at `n_iters=20000` and plot the three loss curves on one figure. One of them will overflow and print runtime warnings; that is the expected result, not a bug in your code. Use a logarithmic y -axis so all three are visible. [5 marks]

Task 4 — Evaluating the fit

[10 marks]

Using the θ from Task 3(b), predict on the *test* set.

- (a) Write `rmse(y_true, y_pred)` and `r2_score_scratch(y_true, y_pred)` from

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_i (y_i - \hat{y}_i)^2}, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

and print both to four decimals. Report R^2 on the *test* set and on the **training** set. [6 marks]

- (b) Plot the residuals $y_i - \hat{y}_i$ against the predicted values $\hat{y}_i$, with a horizontal line at zero. [4 marks]

Keep this plot open during the viva — you will be asked what it does and does not show.

Submission

- One executed notebook, `IC272_Lab4_⟨RollNo⟩.ipynb`, all outputs and plots visible.
- Show it to your TA, who will ask you questions on it. Marks are recorded then.